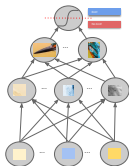
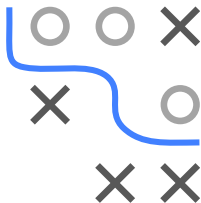


Introduction to Machine Learning

Neural Networks Introduction

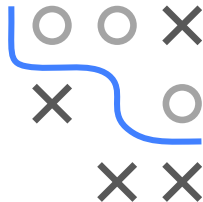
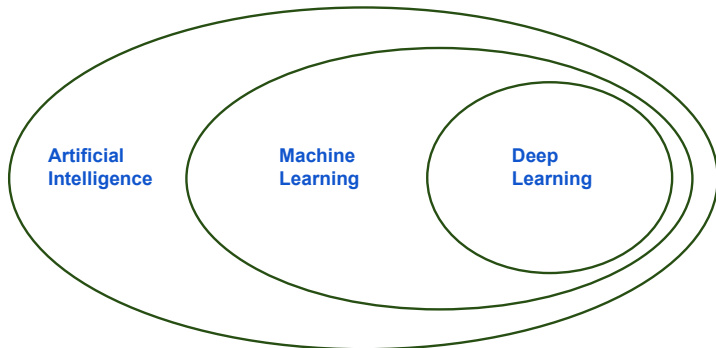


Learning goals

- Relationship of DL and ML
- Concept of representation or feature learning
- Use-cases and data types for DL methods

WHAT IS DEEP LEARNING

- Deep learning is a subfield of ML based on artificial neural networks.



DEEP LEARNING AND NEURAL NETWORKS

Deep learning itself is not *new*:

- Neural networks have been around since the 70s.
- *Deep* neural networks, i.e., networks with multiple hidden layers, are not much younger.

Why everybody is talking about deep learning now:

- Specialized, powerful hardware allows training of huge neural networks to push the state-of-the-art on difficult problems.
- Large amount of data is available.
- Special network architectures for image/text data.
- Better optimization and regularization strategies.

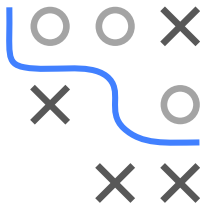


IMAGE CLASSIFICATION WITH NEURAL NETWORKS

“Machine learning algorithms, inspired by the brain, based on learning multiple levels of representation/abstraction.” - Y. Bengio

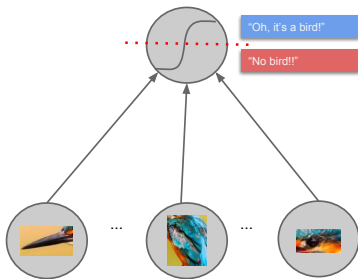
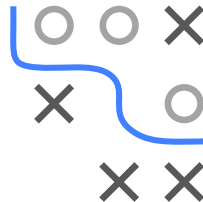


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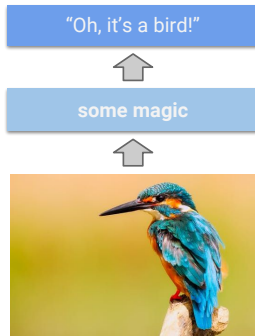
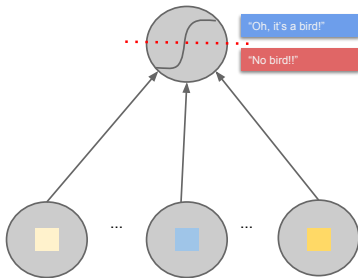
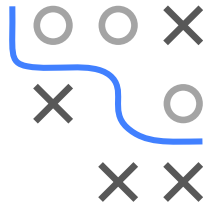


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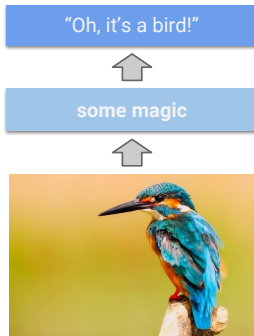
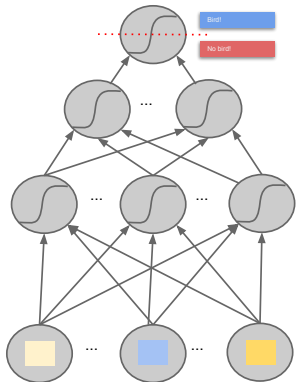
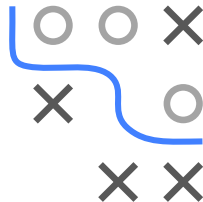
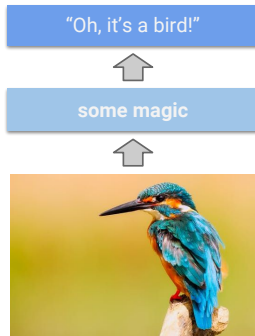
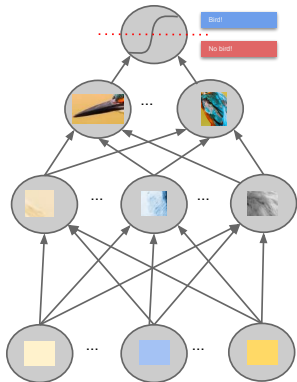
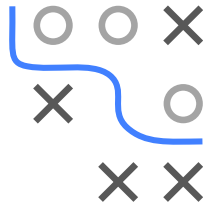


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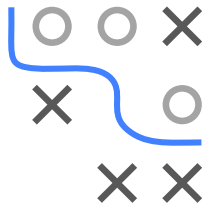
POSSIBLE USE-CASES

Deep learning can be extremely valuable if the data has these properties:

- It is high dimensional.
- Each single feature itself is not very informative but only a combination of them might be.
- There is a large amount of training data.

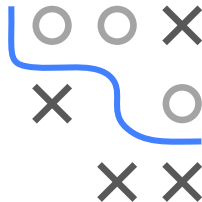
For tabular data, deep learning can make sense in some cases but is interesting from a technical point of view.

- Specialized, powerful hardware allows training of huge neural networks to push the state-of-the-art on difficult problems.
- Without extensive tuning, models like random forests or gradient boosting will often outperform deep learning.
- Deep learning can be valuable for tabular data with categorical features with many levels or complex feature interactions.
- Technical challenges include overfitting, hyperparameter sensitivity, and interpretability.

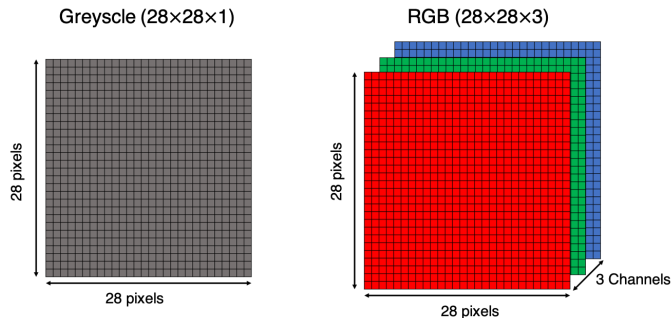


POSSIBLE USE-CASE: IMAGES

- **High Dimensional:** A color image with 255×255 (3 Colors) pixels already has 195075 features.
- **Informative:** A single pixel is not meaningful in itself.
- **Training Data:** Depending on applications huge amounts of data are available.

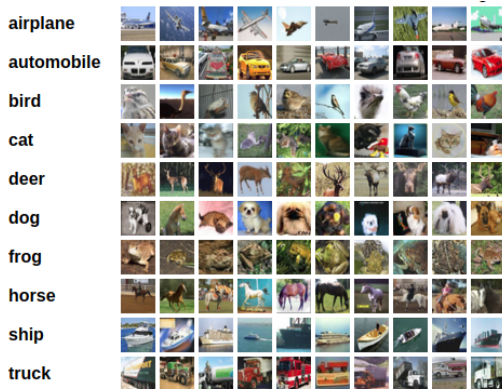


Architecture: **C**onvolutional **N**eural **N**etworks (CNN)



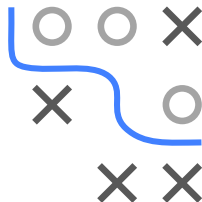
POSSIBLE USE-CASE: IMAGES

- **Image classification** tries to predict a single label for each image.



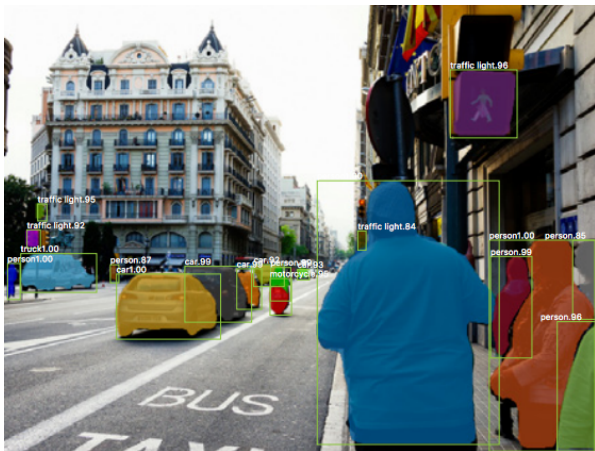
► [Click for source](#)

- CIFAR-10 is a well-known dataset used for image classification. It consists of 60,000 32x32 color images containing one of 10 object classes, with 6000 images per class.



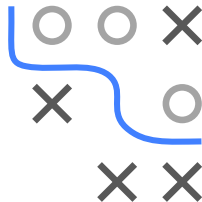
POSSIBLE USE-CASE: IMAGES

- Object Detection



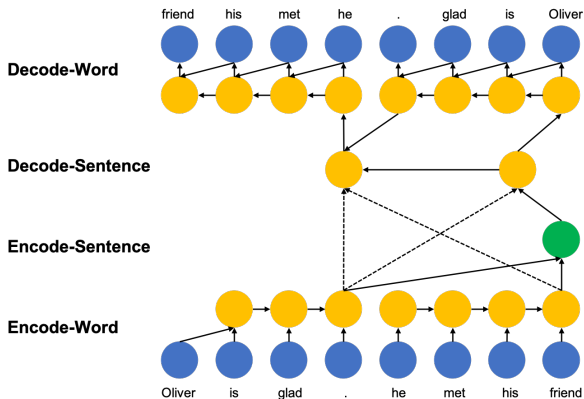
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- Mask R-CNN is a general framework for instance segmentation, that efficiently detects objects in an image while simultaneously generating a high-quality segmentation mask for each instance.

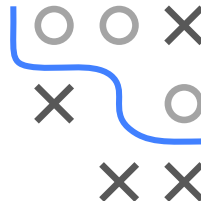


POSSIBLE USE-CASE: TEXT

- **High Dimensional:** Each word can be a single feature (300000 words in the German language).
- **Informative:** A single word does not provide much context.
- **Training Data:** Huge amounts of text data available.



Architecture: **R**ecurrent **N**eural **N**etworks (RNN)



POSSIBLE USE-CASE: TEXT CLASSIFICATION

- **Sentiment Analysis** is the application of natural language processing to systematically identify the emotional and subjective information in texts.



Positive

Great job! Your customer support is fantastic.



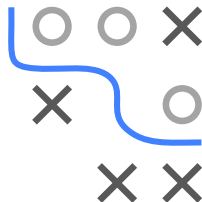
Neutral

Not bad, but it should be improved in the future.



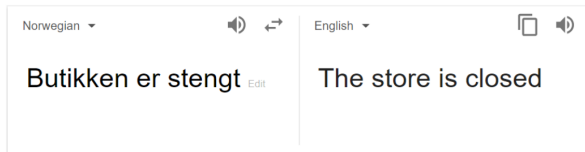
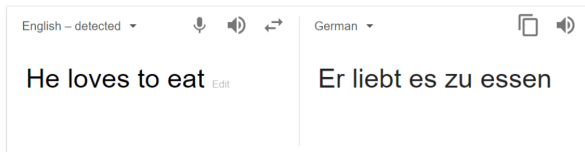
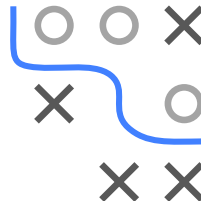
Negative

The worst customer service I have ever seen.



POSSIBLE USE-CASE: TEXT

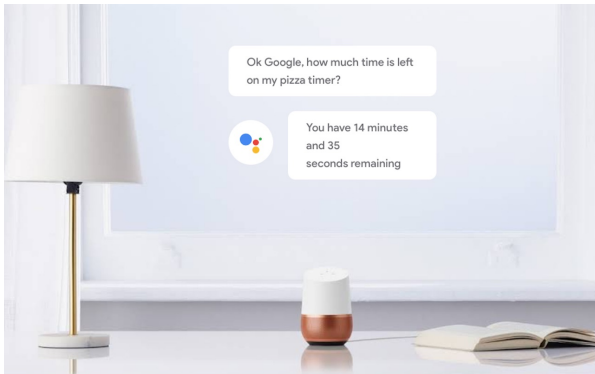
- **Machine Translation** (e.g. google translate) Neural machine translation exploits neural networks to predict the likelihood of a sequence of words, typically modeling entire sentences in a single integrated model.



APPLICATIONS OF DEEP LEARNING: SPEECH

- **Speech Recognition and Generation** (e.g. google assistant)

Neural network extracts features from audio data for downstream tasks, e.g., to classify emotions in speech.



► [Click for source](#)

